

HybridAgent

The On-Chain Escrow Marketplace for Property & Vehicle Sales

Slide 1 — The Problem

Real estate and vehicle deals involve three parties: **an owner, a buyer, and an agent**.

The agent finds the buyer, negotiates the deal, and does all the work — but they get paid **last**, if at all.

What goes wrong today

- The buyer pays the seller directly (cash, bank transfer).
- The seller holds the full amount.
- The agent chases their commission — sometimes for months.
- The seller can simply refuse to pay. There is no enforceable mechanism.

Agents lose commissions every day because trust breaks down at the moment money moves.

Slide 2 — The Solution

HybridAgent is a marketplace where the buyer's payment settles fully **on-chain in USDC** and the agent's commission is carved out **atomically in the same transaction** — before a single dollar reaches the seller.

The agent cannot be cheated because the money never touches the seller's wallet until the agent's share has already been sent.

What makes it work

- A smart contract holds the buyer's full payment in escrow.
 - The agreed commission is locked in a signed on-chain mandate before the deal opens.
 - On completion, one transaction pays **agent** → **commission, platform** → **fee, seller** → **the rest** — simultaneously.
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Slide 3 — Who Is It For

Role	Who they are
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Owner / Seller	A person or company selling a property or vehicle
Agent / Broker	A licensed agent or independent broker representing the seller
Buyer	Anyone purchasing a listed asset

A user can be both an owner and an agent — the role is **per-listing**, not per-account.

Slide 4 — Two Listing Types

1. Owner Direct

- The owner lists their **own** asset.
- No agent is involved.
- **Commission = 0%.**
- Buyer pays → escrow holds → seller confirms → funds released directly to seller.

2. Agent Brokered

- An agent lists a property or vehicle on **behalf of an owner**.
 - Owner and agent agree on a commission rate (up to **30%**) and sign a **mandate on-chain**.
 - When the deal closes, the commission is paid to the agent automatically — no chase, no delay.
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Slide 5 — How to Use the Platform (What's Live Today)

Step 1 — Register

- Create an account with your name, email, and phone number.
- Complete KYC (identity verification) to unlock transacting features.

Step 2 — Sign In with Your Email Wallet

- Sign in via email — a secure embedded wallet is created automatically using **Privy**.
- No browser extension needed. No seed phrase to manage.
- Your wallet address is shown in the navbar once connected.

Step 3 — Browse Listings

- View all open properties and vehicles on the Listings page.
- Each listing shows the price in USDC, commission rate, and agent details.
- A live **price quote** breaks down exactly what the agent earns and what the seller receives.

Step 4 — Create a Listing (Agents)

- Choose listing type: **Owner Direct** or **Agent Brokered**.
- Add title, asset type, price in USDC, and photos.
- For agent-brokered: enter the owner's name and email.

The owner receives an email notification that their asset was listed and a wallet was reserved for their proceeds.

Step 5 — On-Chain Settlement (*contracts deployed, UI integration in progress*)

- The buyer funds the full sale price into the **HybridEscrow** smart contract.
 - A 7-day dispute window opens.
 - On confirmation, one transaction splits the funds: agent gets commission, platform gets fee, seller gets the rest.
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Slide 6 — How Everyone Gets Paid

When a deal settles, one atomic transaction distributes the funds:

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Buyer pays:  $100,000 USDC (full price)
Agent gets:   $5,000    (5% commission – agreed in mandate)
Platform gets: $2,000   (2% platform fee)
Seller gets:  $93,000   (remainder – price minus commission and fee)
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Caps enforced by the smart contract

- Maximum agent commission: **30%**
- Maximum platform fee: **10%**
- Commission + fee combined cannot exceed **100%** of the sale price.

No one is waiting. No one is chasing. The contract does it in one block.

Slide 7 — What If Something Goes Wrong?

Dispute Resolution

- Either party can raise a dispute **within 7 days** of the buyer funding escrow.
- A neutral **arbiter** reviews the case.
- The arbiter has two options:
 1. **Release** — settle normally (agent + platform + seller get paid).
 2. **Refund** — return the full amount to the buyer.

Seller Timeout Claim

- If the buyer funds the deal but never confirms and never disputes, the **seller can claim the funds** after the 7-day window closes automatically.
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Slide 8 — The Mandate: Agent Protection Built In

Before any deal opens, the agent's commission is locked in a **MandateRegistry** contract.

Step	Who acts	What happens
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1	Owner	Creates a mandate: assigns agent, sets commission %, sets expiry
2	Agent	Accepts the mandate on-chain
3	Deal opens	Commission rate is snapshotted into the escrow deal
4	Deal closes	Commission is paid from that snapshot — owner cannot change it

The owner can revoke a mandate before it is consumed by a deal, but **not after**.

Slide 9 — What's Built vs What's Next

Feature	Status
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User registration + KYC-gated routes	Live
Email login + embedded wallet (Privy)	Live
Listing creation — owner direct & agent brokered	Live
Owner email notification on listing creation	Live
Browse listings + listing detail page	Live
Price quote (commission split preview)	Live
Leaderboard + agent reviews	Live
HybridEscrow smart contract (Sepolia)	Deployed
MandateRegistry smart contract (Sepolia)	Deployed
On-chain buy flow (buyer funds escrow via UI)	In progress
On-chain mandate signing via UI	In progress
Deal confirmation + dispute via UI	In progress

Slide 10 — Technology Stack

Layer	Technology
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Frontend	Next.js 15 (App Router), React 19, Tailwind CSS v4
Wallet / Auth	Privy — email login, embedded smart wallet, no extension needed
Smart Contracts	Solidity 0.8.24, Hardhat, OpenZeppelin — deployed on Ethereum Sepolia
Backend API	Node.js, Express, PostgreSQL (Railway)
Payments	USDC (ERC-20 stablecoin) on Ethereum Sepolia
File Storage	Cloudinary (listing images)
Email	Resend — transactional notifications
Indexer	ethers v6, polling on-chain events into Postgres for fast reads

Slide 11 — Summary

The old way	HybridAgent
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Agent trusts the seller to pay	Commission locked in smart contract before deal opens
Payment by bank transfer — manual, slow	USDC escrow — instant, on-chain, auditable
Disputes handled by lawyers	7-day window + neutral arbiter on-chain
Agent chases commission for months	Agent paid in the same block the deal closes
No transparency on commission	Commission rate visible and immutable on-chain

HybridAgent makes agent commission theft structurally impossible.

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